EAST CRAZIES EXAMPLE SOILS SCOPE OF WORK

Date: 2026-05-07

Audience: WLG demonstration package
Purpose: show that the system can produce resource-area scopes of work for a new NEPA project from a structured project intake
Example project: East Crazy Inspiration Divide land exchange
Forest: Custer Gallatin National Forest
District: Bozeman Ranger District
Resource area: Soils

This example assumes a new project is in early planning, before late-stage NEPA materials or resource analysis packages exist. The SOW is generated from the proposed action, project location, preliminary action elements, parcel/trail/access features, likely resource triggers, and standard public source requirements.

This is the capability WLG should see: the system can translate the controlling source stack into a tight specialist-report SOW. For this soils example, the stack includes NEPA, current USDA NEPA procedures, NFMA/project-consistency requirements, Forest Service handbook and manual direction, Region 1 soil management direction, the applicable Forest Plan, soil-quality monitoring methods, National BMPs, and official soil-survey data. The SOW turns those sources into fieldwork, analysis, deliverables, and acceptance criteria that should produce a defensible soils specialist report without pretending the planning package has already made a NEPA or legal sufficiency determination.

For the East Crazies example, the soils scope is driven by:

- proposed new or rerouted trail segments;
- trailhead improvements;
- stream crossings, wetland/riparian approaches, wet soils, hydric soils, erosion, and sediment delivery;
- parcel restrictions, access/easements, grazing-related improvements, and acquired-parcel soil management needs.

The consultant product is a Soils Specialist Report or, if approved by the Forest Service soils reviewer, a short issue-screening memo. Either product must support later EA drafting, Forest Plan consistency documentation, mitigation commitments, and implementation planning.

This is a planning and contracting support SOW. It is not a final agency decision, legal sufficiency review, or NEPA compliance determination.

FIELDWORK

Before mobilization, prepare a survey-trigger table that ties each proposed action element to the source that requires field verification. Do not include generalized reconnaissance as a required task. If a survey below is not triggered, record the reason in the table.

1. Soil Survey And Activity-Area Delineation

Required for each ground-disturbing action area: proposed trail construction or reroute, trailhead work, access/staging, restoration work, crossing approaches, and acquired-parcel management areas where future NFS management could disturb soil.

- Use NRCS Web Soil Survey/SSURGO, DEM slope data, NHD/NWI, wetland/riparian layers, aerial imagery, and Forest Plan soil/watershed direction to map the soil map units and activity areas before the field visit.
- Field-verify soil map unit boundaries, slope class, landform, drainage position, hydric or wet soil indicators, erosion hazard, revegetation limits, and sediment connectivity where the action could change soil condition or water quality.
- Delineate the activity area used for Forest Plan FW-STD-SOIL-01 and MON-SOIL-01. For linear trail features, define the width, length, acreage, and rationale used to translate the trail footprint into an activity area.

2. Detrimental Soil Disturbance Survey

Required where a management activity could create or add to detrimental soil conditions in an activity area. This is the core East Crazies soils survey for trail construction or reroute, trailhead work, access/staging, restoration, and any disposal or acquisition parcel work that changes soil condition.

- Use the Forest Service Soil-Disturbance Field Guide and Forest Soil Disturbance Monitoring Protocol as the field method for classifying existing soil disturbance.
- Document compaction, rutting, displacement, puddling, surface erosion, severe burning if present, organic-matter loss, and mass movement.
- Estimate percent detrimental soil disturbance by activity area, including existing disturbance. The report must show whether the proposed action would exceed, avoid, or require restoration under the Custer Gallatin Forest Plan FW-STD-SOIL-01 threshold.
- Identify where MON-SOIL-01 pre-implementation and post-implementation monitoring is needed. If the Forest Service accepts a linear-feature substitute metric, document the metric and why it remains comparable to the Forest Plan disturbance threshold.

3. CWD And Organic-Substrate Survey

Required only if the proposed action includes vegetation activity areas, burn piles, slash treatment, forest-floor removal, revegetation/restoration, or other activity that would affect coarse woody debris, litter, duff, or organic soil cover.

- If triggered, estimate percent coarse woody debris in vegetation activity areas for MON-SOIL-02 and document litter/duff, ground cover, organic matter, and restoration substrate conditions.
- For trailhead, access, or restoration work that removes organic cover but is not a vegetation activity area, record the affected area, expected organic-cover loss, and restoration requirement.
- If no vegetation activity area or organic-substrate effect is present, mark this survey as not triggered.

4. Slope, Equipment, And Unstable-Area Survey

Required where mechanized construction, ground-based equipment, temporary access, blading, scarification, pits/trenches, burn piles, or steep/unstable landforms are proposed.

- Verify sustained grades over 40 percent where ground-based equipment could be used.
- Identify high mass-wasting potential, unstable slopes, wet soils, hydric soils, sensitive soil map units, and areas where equipment exclusion or seasonal restriction is needed.
- Record locations where temporary road blading, mechanical scarification, soil pits/trenches, or burn pile restoration would trigger Forest Plan soil guidelines FW-GDL-SOIL-01 through FW-GDL-SOIL-08.

5. Waterbody Interface And BMP Survey

Required where trail, trailhead, access, crossing, drainage, or restoration work is near streams, wetlands, riparian areas, seeps, springs, swales, drainage paths, or other sediment-routing features.

- Field-check proposed stream and wetland approaches, crossing locations, drainage outlets, slope breaks, toe slopes, and hydrologic connectivity to waterbodies.
- Identify site-specific National BMP needs: drainage spacing and outlet protection, crossing hardening, sediment control, wet-weather limits, revegetation, decompaction, and construction timing restrictions.
- Record whether the Inspiration Divide Trail reroute would remove or reduce an existing stream/wetland crossing effect and what design criteria are needed for the replacement alignment.

6. Minimum Survey Record

For each triggered survey location or activity area, record:

- date, observer, weather, soil moisture, access limits, survey trigger, and source driver;
- activity-area ID, action element, soil map unit, GPS identifier, photo IDs, and field form ID;

- protocol used, measurements taken, disturbance class, percent disturbance where required, and linear-feature substitute metric if approved;
- slope, landform, drainage position, hydric/wet soil indicators, unstable-area indicators, and sediment-connectivity notes;
- BMP, design criterion, mitigation, restoration, monitoring requirement, data gap, or revisit need.

Conduct fieldwork during snow-free conditions when the mineral soil surface, organic cover, and drainage paths can be observed. Conduct a wet-season or post-storm revisit only if waterbody connectivity, runoff, erosion, or sediment delivery cannot be determined during the main survey.

ANALYSIS

Use Appendix A as the authority-to-report control table. The analysis should answer only the soil questions triggered by the proposed action and should tie each conclusion to field evidence, GIS data, cited source direction, or a documented Forest Service assumption.

1. Existing Condition

Summarize the soil setting for each affected activity area:

- soil map unit, slope, landform, drainage, hydric/wet soil indicator, erosion hazard, and revegetation or compaction limitation;
- existing detrimental soil disturbance, route drainage, erosion, sediment connectivity, and restoration need;
- CWD, litter/duff, ground cover, or organic-substrate condition only where the CWD/organic-substrate survey is triggered;
- disposal-parcel and acquisition-parcel soil conditions only where parcel restrictions or future NFS management create a soil issue;
- data gaps that limit effects conclusions.

2. Effects Indicators

Use the same indicators for no action, proposed action, and any Forest Service-provided alternative:

- activity-area acreage for area-based work;
- length, width, crossing count, drainage feature count, wetland/riparian approach count, and representative disturbance class for linear trail work;
- percent detrimental soil disturbance where FW-STD-SOIL-01 applies;
- hydrologic connectivity, erosion risk, revegetation feasibility, wet-soil limitation, unstable-area risk, and restoration need.

Do not force linear trail effects into acreage-only conclusions. Use acreage where it is useful and linear metrics where they better describe the effect.

3. Proposed-Action Effects

Analyze soil effects from the proposed trail construction, trail reroute, trailhead work, crossings, access/staging, restoration, and parcel-management changes. Address whether the action would:

- create, add to, or restore detrimental soil disturbance;
- route sediment toward streams, wetlands, riparian areas, seeps, springs, swales, or drainage paths;
- reduce an existing stream/wetland crossing effect through the Inspiration Divide Trail reroute;
- require erosion, compaction, drainage, topsoil, revegetation, wet-weather, or equipment controls;
- create soil-management obligations on acquired NFS parcels or reduce soil risk through parcel restrictions after conveyance.

4. Forest Plan Consistency

Use Appendix A to document the Forest Plan consistency finding needed for the soils report. The report must state:

- whether FW-STD-SOIL-01 applies, is met, requires restoration, or is not triggered;
- which FW-GDL-SOIL-01 through FW-GDL-SOIL-08 guidelines apply and what design feature or equivalent measure satisfies each one;
- whether MON-SOIL-01 or MON-SOIL-02 monitoring is required or screened out;
- which related road/trail, watershed, riparian, wetland, revegetation, invasive-species, backcountry, or recommended-wilderness components change the soils analysis.

5. Mitigation, BMPs, And Monitoring

Provide one mitigation/BMP/monitoring matrix. Each row must include action element, location, triggering source, soil concern, required measure, timing, responsible party, and verification method. Include only measures tied to a triggered source or a site-specific field finding.

6. Cumulative Effects And Uncertainty

Evaluate cumulative soil effects only where a cause-and-effect relationship exists. Use the same indicators listed above and consider existing/proposed routes, trailheads, unauthorized routes, drainage, grazing, wetland/riparian disturbance, foreseeable trail or road work, private land development risk after conveyance, and vegetation/restoration actions.

Close the analysis with unresolved assumptions, field limitations, missing design details, and Forest Service decisions needed before the soils conclusions can support later NEPA or implementation files.

DELIVERABLES

1. Kickoff And Data Request Log

Provide kickoff notes, received and missing data, access constraints, source materials used, stop conditions, and Forest Service decisions needed before fieldwork.

Acceptance criteria:

- identifies missing GIS, trail design, parcel, wetland, route, construction-method, or Forest Plan data;
- identifies the current laws, regulations, Forest Service handbook/manual direction, Region 1 direction, Forest Plan components, BMPs, and soil-data sources that will govern the soils work;
- identifies whether the work will proceed as a standalone soils report or issue-screening memo;
- identifies field-season and access constraints.

2. Fieldwork Plan

Provide field objectives, activity-area definitions, field maps, observation methods, field forms, data conventions, schedule, safety/access plan, and coordination needs.

Acceptance criteria:

- covers the proposed trail, reroute, trailhead, crossings, wetland/riparian interfaces, parcel restrictions, and other project-specific soil trigger areas;
- explains how soil disturbance will be documented for linear features;
- identifies whether wet-season or post-storm verification may be needed.

3. Field Data Package

Provide GPS data, completed field forms, photo log, soil-check notes, soil map unit verification, wetland/riparian and sediment-connectivity notes, and a short field findings memo.

Acceptance criteria:

- every material field conclusion is traceable to location, date, observer, photo, and activity area;
- sensitive soil, hydric/wet soil, erosion, compaction, rutting, displacement, drainage, and sediment concerns are mapped or explicitly screened out;
- data are suitable for Custer Gallatin GIS and ID-team use.

4. Analysis Tables And Map Atlas

Provide:

- soil map unit and activity-area crosswalk;
- existing and expected disturbance table;
- action-element by soil concern matrix;
- stream/wetland crossing and sediment-connectivity table;
- mitigation/BMP/monitoring matrix;
- Appendix A authority-to-report crosswalk with project-specific updates;
- Forest Plan requirements matrix;
- data-gap and uncertainty table;
- project, soil-analysis area, soil map unit, slope/erosion hazard, wetland/riparian, road/trail, observation, and mitigation maps.

Acceptance criteria:

- each fieldwork, analysis, mitigation, and monitoring item is tied to its controlling source;
- maps include scale, north arrow, coordinate system, data date, project boundary, activity-area IDs, and source attribution;
- tables distinguish no action, proposed action, preliminary alternatives, and any later analyzed alternative;
- linear-feature effects use useful measures and do not rely on misleading acreage-only metrics.

5. Draft Soils Specialist Report Or Issue-Screening Memo

Provide a draft report or approved issue-screening memo with affected environment, methods, spatial and temporal scale, resource indicators, direct and indirect effects, cumulative effects, mitigation/BMPs/monitoring, Forest Plan requirements, references, and appendices.

Acceptance criteria:

- directly addresses the proposed action and each soil-triggering action element;
- states whether FW-STD-SOIL-01 applies or is screened out for trail work;
- ties conclusions to Appendix A, field evidence, GIS evidence, official data, or specialist judgment.

6. EA-Ready Soils Text

Provide concise text for the future EA or appendix covering affected environment, environmental consequences, mitigation/design criteria, Forest Plan requirements, and cumulative effects if needed.

Acceptance criteria:

- suitable for public disclosure;
- references the soils report or issue-screening memo instead of duplicating all technical detail;
- identifies design criteria or monitoring that must carry into later decision or implementation files.

7. Final Report And Data Package

Provide final report or memo, final PDF, editable source file, comment-response log, final tables, map atlas, GIS layers or geodatabase/geopackage, photo log, field forms, source data inventory, metadata/readme, and data limitations memo.

Acceptance criteria:

- resolves Forest Service and ID-team comments;
- preserves source citations and field-data traceability;
- preserves the authority-to-report crosswalk so the final specialist report remains defensible;
- documents residual uncertainty and remaining Forest Service decisions;
- final PDF starts with a valid ‘%PDF-‘ header.

APPENDIX A. AUTHORITY-TO-REPORT CROSSWALK

Use this appendix as the starting control table for the soils specialist report. Update it only if Forest Service direction, source currentness, or the proposed action changes.

A1. NEPA, USDA NEPA Procedures, And FSH 1909.15

- Source: NEPA; USDA NEPA procedures at 7 CFR part 1b; FSH 1909.15.
- Trigger: Federal land exchange, trail, trailhead, access, and parcel-management decisions that need soils effects support for an environmental assessment or issue-screening memo.
- Report output: concise affected environment, methods, direct and indirect effects, cumulative effects if useful, mitigation/monitoring, source citations, and data limitations.
- Acceptance test: each conclusion is traceable to reliable existing data, field evidence, or a documented Forest Service assumption; the report does not make the agency decision.

A2. Third-Party Or Consultant-Prepared Analysis

- Source: 7 CFR part 1b contractor/third-party preparation procedures and Forest Service direction.
- Trigger: WLG or a consultant prepares soils analysis for Forest Service review.
- Report output: editable report, field forms, GIS/map package, photo log, source inventory, assumptions, limitations, and comment-response log.
- Acceptance test: Forest Service can independently review the methods, source citations, field evidence, and assumptions before using the report.

A3. NFMA And 36 CFR Part 219

- Source: NFMA; 36 CFR part 219, especially project/activity consistency with the applicable Forest Plan.
- Trigger: Any NFS project or authorization after approval of the Custer Gallatin Forest Plan.
- Report output: Forest Plan consistency section that addresses applicable desired conditions, standards, guidelines, suitability, and monitoring components.
- Acceptance test: the report states whether the action complies with standards, follows guidelines or uses an equally effective design, and avoids foreclosing applicable desired conditions.

A4. Custer Gallatin Forest Plan Soil Desired Conditions

- Source: FW-DC-SOIL-01, FW-DC-SOIL-02, and FW-DC-SOIL-03.
- Trigger: Trail, trailhead, crossing, access, restoration, and parcel-management actions that could affect soil productivity, hydrologic function, erosion, or ground cover.
- Report output: existing-condition and effects discussion for soil productivity, erosion, sediment connectivity, infiltration, ground cover, and restoration feasibility.
- Acceptance test: the report explains how the action maintains or moves toward the soil desired conditions, or identifies the design changes needed.

A5. Custer Gallatin Forest Plan Soil Standard

- Source: FW-STD-SOIL-01.
- Trigger: Any management activity that could create or add to detrimental soil conditions in an activity area.
- Report output: activity-area delineation, existing DSD estimate, expected DSD estimate, linear-trail metric rationale if used, and restoration/design measures needed to avoid exceeding the standard.
- Acceptance test: the report states whether FW-STD-SOIL-01 applies, is met, requires restoration, or is not triggered.

A6. Custer Gallatin Forest Plan Soil Guidelines

- Source: FW-GDL-SOIL-01 through FW-GDL-SOIL-08.
- Trigger: Ground-based equipment, temporary access, blading, scarification, pits/trenches, burn piles, steep slopes, high mass-wasting potential, CWD, or restoration work.
- Report output: guideline applicability table with action element, location, field finding, design feature, mitigation, or equally effective alternative.
- Acceptance test: every triggered guideline is carried into the design criteria or explicitly screened out.

A7. Custer Gallatin Forest Plan Soil Monitoring

- Source: MON-SOIL-01 and MON-SOIL-02.
- Trigger: DSD monitoring where FW-STD-SOIL-01 applies; CWD monitoring where vegetation activity areas or organic-substrate effects are present.
- Report output: monitoring recommendation with location, metric, timing, responsible party, and success criterion.
- Acceptance test: the report identifies pre/post DSD monitoring and CWD monitoring where triggered, or states why they are not triggered.

A8. Region 1 Soil Management Direction

- Source: USDA Forest Service Manual, Region 1 Supplement, FSM 2550 Soil Management.
- Trigger: Forest Service management activity that could affect soil productivity or create detrimental soil disturbance.
- Report output: regional soil-quality method, DSD threshold logic, productivity-risk discussion, and restoration requirements.
- Acceptance test: regional soil direction is visible in the methods, effects indicators, and mitigation/monitoring matrix.

A9. Soil-Disturbance Field Guide And Monitoring Protocol

- Source: Forest Service Soil-Disturbance Field Guide and Forest Soil Disturbance Monitoring Protocol.
- Trigger: Any required DSD survey or soil-disturbance classification.
- Report output: disturbance classes, field forms, photo/GPS evidence, activity-area summary, and protocol deviations.
- Acceptance test: disturbance conclusions are repeatable from the field record.

A10. NRCS Web Soil Survey And SSURGO

- Source: NRCS Web Soil Survey and SSURGO.
- Trigger: Every mapped activity area and any parcel where soil limitations affect the proposed action.
- Report output: soil map unit/activity-area crosswalk, limitation ratings, hydric/wet soil indicators, erosion hazard, drainage, revegetation limits, and data limitations.
- Acceptance test: official soil data are cited and field-verified where the action creates a soil issue.

A11. National BMPs

- Source: Forest Service National Best Management Practices Program.
- Trigger: Trail, trailhead, access, crossing, drainage, or restoration work with water-quality or

sediment-delivery risk.
- Report output: BMP matrix for drainage control, hydrologic disconnection, crossing hardening, sediment control, wet-weather restrictions, decompaction, revegetation, and inspection.
- Acceptance test: BMPs are tied to mapped locations and verification methods, not listed generically.

A12. East Crazies Example Action Elements

- Source: East Crazies structured intake assumptions.
- Trigger: Sweet Trunk Trail work, Inspiration Divide Trail reroute, Big Timber Canyon Trailhead improvements, stream/wetland approaches, access/easement features, parcel restrictions, grazing improvements, and acquired-parcel management needs.
- Report output: action-element by soil-concern matrix, field-survey trigger table, effects indicators, maps, mitigation/BMP/monitoring matrix, and issue-screening rationale if a full report is not required.
- Acceptance test: every soils conclusion is tied to an action element or explicitly screened out.

SOURCES CITED

- National Environmental Policy Act, 42 U.S.C. 4321 et seq.
- National Forest Management Act, 16 U.S.C. 1600 et seq.
- USDA NEPA procedures, 7 CFR part 1b:
  https://www.ecfr.gov/current/title-7/subtitle-A/part-1b
- National Forest System land management planning, 36 CFR part 219:
  https://www.ecfr.gov/current/title-36/chapter-II/part-219
- Forest Service Handbook (FSH) 1909.15, National Environmental Policy Act Handbook:
  https://www.fs.usda.gov/cgi-bin/Directives/get_dirs/fsh?1909.15=
- USDA Forest Service Manual, Region 1 Supplement, FSM 2550 Soil Management.
- Custer Gallatin National Forest Land Management Plan:
  https://www.fs.usda.gov/r01/custergallatin/planning/forest-plan/custer-gallatin-land-management-plan-forest-plan-revision
- Forest Service Soil Quality Monitoring resources:
  https://www.fs.usda.gov/soils/monitoring.shtml
- Forest Service Soil-Disturbance Field Guide:
  https://www.fs.usda.gov/t-d/php/library_card.php?p_num=0819+1815P
  https://www.fs.usda.gov/rm/pubs_other/rmrs_2009_napper_c001.pdf
- NRCS Web Soil Survey and SSURGO:
  https://www.nrcs.usda.gov/resources/data-and-reports/web-soil-survey
- Forest Service National Best Management Practices Program:
  https://www.fs.usda.gov/naturalresources/watershed/bmp.shtml
- East Crazies example project intake assumptions: land exchange, proposed trail construction, proposed trail reroute, trailhead improvements, parcel restrictions, easements, wetland/riparian interfaces, grazing-related improvements, and acquired-parcel management needs.